

# Term 3 Practice Activity: C6 Standardisation in Context

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## Evaluated criteria

- C5. Uses the normal standard distribution and its tables to calculate probabilities in contextual situations.

## Contents

## Evaluated criteria

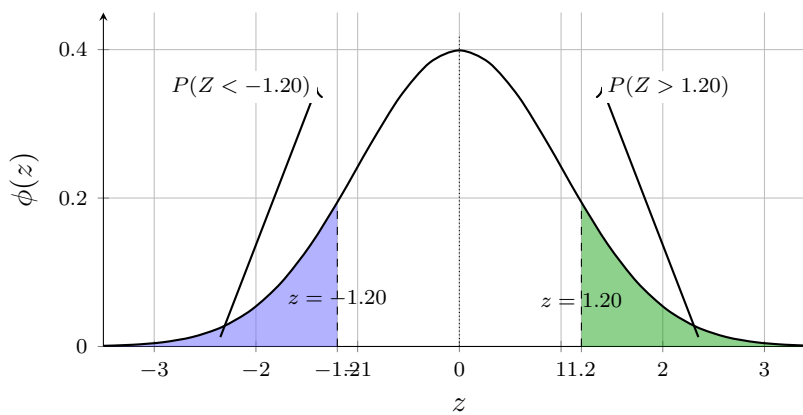
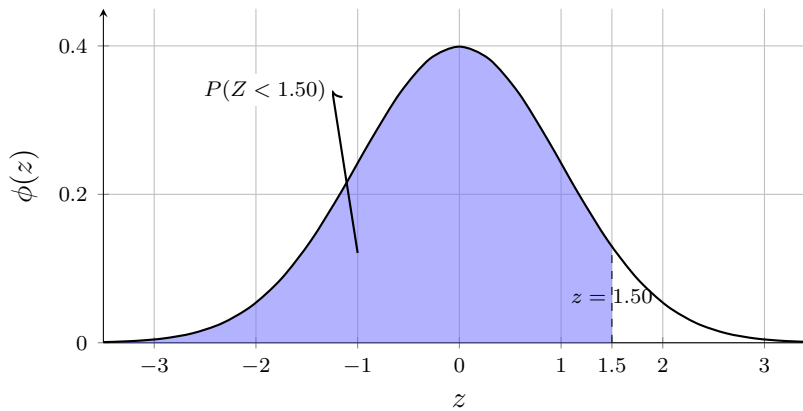
- C6.

Standard normal table (values of  $\Phi(z)$ ).

| $z$ | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |

## 1 Problem 1: Normal Distribution Left Tail with Positive and Negative z-values

**Problem.** Exam scores follow  $X \sim N(70, 8^2)$ . Find  $P(X < 82)$  and  $P(X < 60.4)$ .



$$Z = \frac{X - 70}{8}.$$

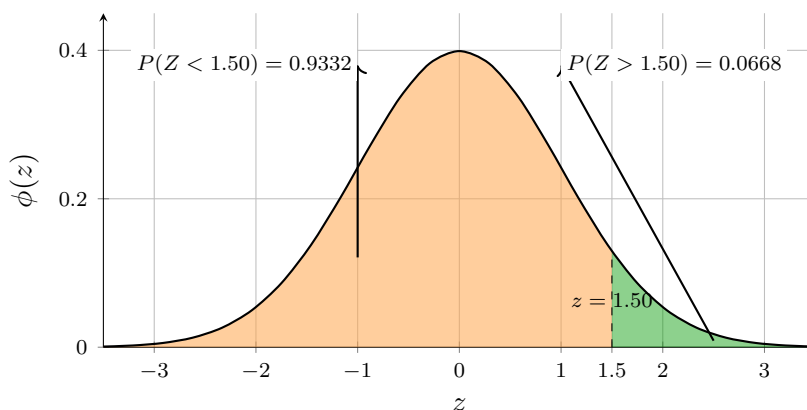
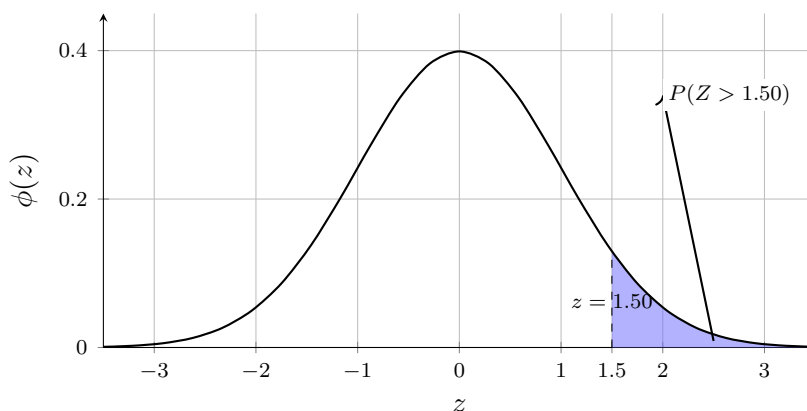
$$P(X < 82) = P\left(Z < \frac{82 - 70}{8}\right) = P(Z < 1.50) = 0.9332.$$

$$P(X < 60.4) = P\left(Z < \frac{60.4 - 70}{8}\right) = P(Z < -1.20) = 1 - 0.8849 = 0.1151.$$

**Interpretation.** About 93.32% score below 82, while 11.51% score below 60.4.

## 2 Problem 2: Normal Distribution Right Tail with Positive and Negative z-values

**Problem.** Delivery times follow  $X \sim N(56, 5^2)$  minutes. Find  $P(X > 63.5)$  and  $P(X > 52)$ .



$$Z = \frac{X - 56}{5}.$$

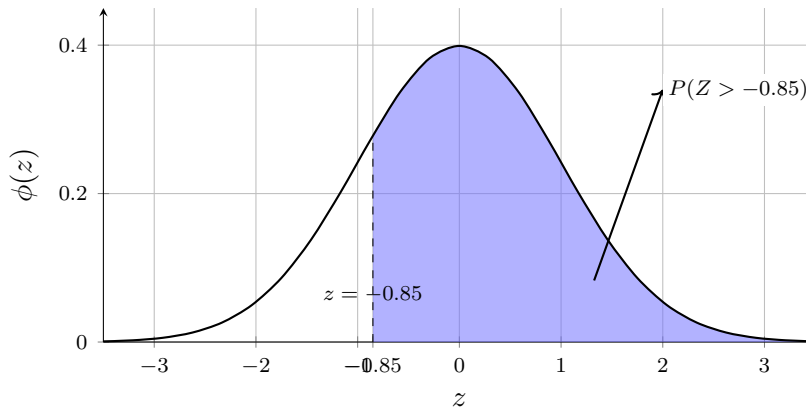
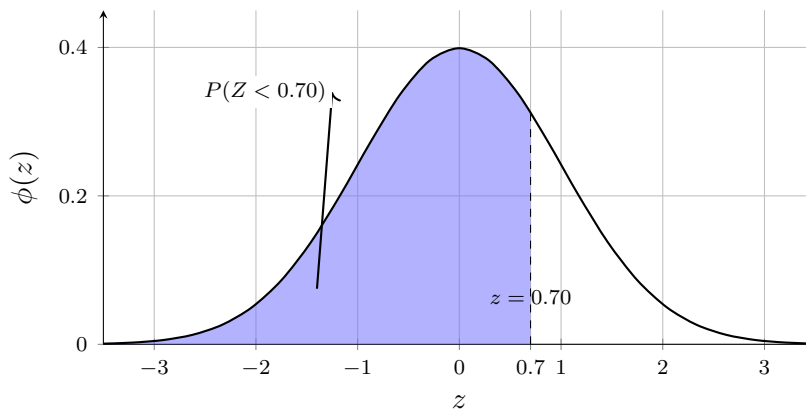
$$P(X > 63.5) = P\left(Z > \frac{63.5 - 56}{5}\right) = P(Z > 1.50) = 1 - 0.9332 = 0.0668.$$

$$P(X > 52) = P\left(Z > \frac{52 - 56}{5}\right) = P(Z > -0.80) = \Phi(0.80) = 0.7881.$$

**Interpretation.** 6.68% take more than 63.5 min, and 78.81% take more than 52 min.

## 3 Problem 3: Normal Distribution Mixed Left-Tail and Right-Tail Reasoning

**Problem.** Production weights follow  $X \sim N(500, 20^2)$  g. Find  $P(X < 514)$  and  $P(X > 483)$ .



$$Z = \frac{X - 500}{20}.$$

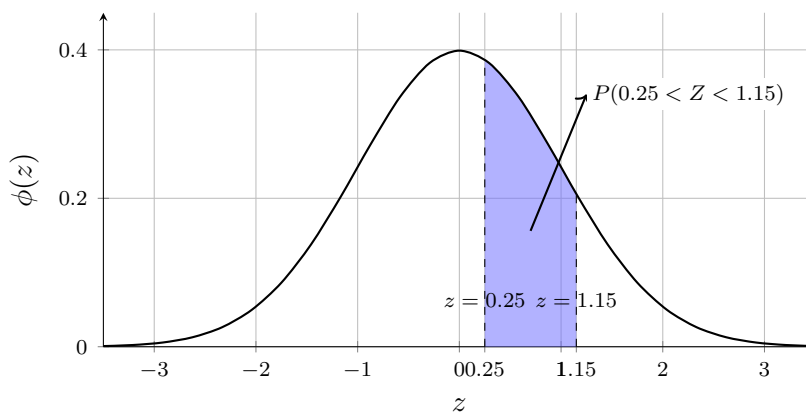
$$P(X < 514) = P\left(Z < \frac{514 - 500}{20}\right) = P(Z < 0.70) = 0.7580.$$

$$P(X > 483) = P\left(Z > \frac{483 - 500}{20}\right) = P(Z > -0.85) = \Phi(0.85) = 0.8023.$$

**Interpretation.** 75.80% are below 514 g, and 80.23% are above 483 g.

#### 4 Problem 4: Normal Distribution Central Band with Positive z-bounds

**Problem.** Weekly app usage follows  $X \sim N(14, 4^2)$  hours. Find  $P(15 < X < 18.6)$ .



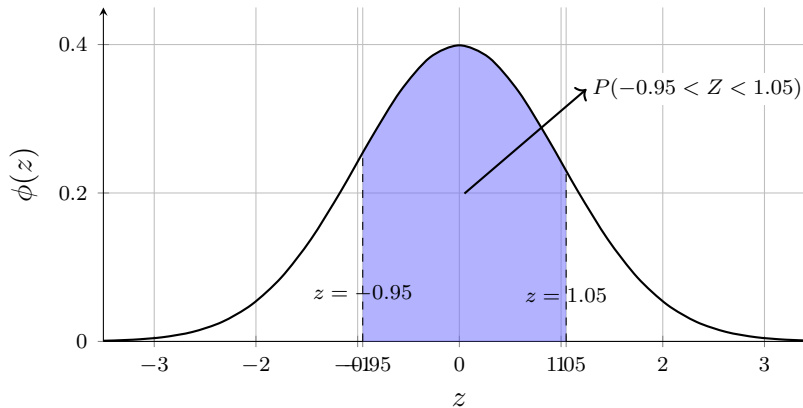
$$Z = \frac{X - 14}{4}, \quad P(15 < X < 18.6) = P(0.25 < Z < 1.15).$$

$$P = \Phi(1.15) - \Phi(0.25) = 0.8749 - 0.5987 = 0.2762.$$

**Interpretation.** 27.62% of students use the app between 15 h and 18.6 h weekly.

## 5 Problem 5: Normal Distribution Central Band with One Negative and One Positive z-bound

**Problem.** Commute times follow  $X \sim N(40, 10^2)$  minutes. Find  $P(30.5 < X < 50.5)$ .



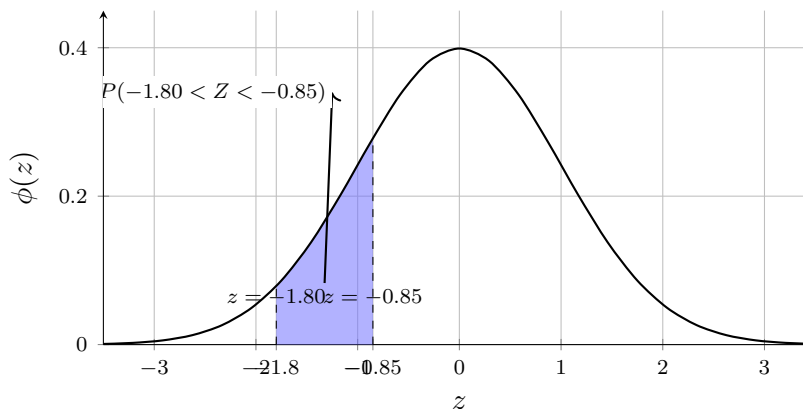
$$Z = \frac{X - 40}{10}, \quad P(30.5 < X < 50.5) = P(-0.95 < Z < 1.05).$$

$$P = \Phi(1.05) - \Phi(-0.95) = 0.8531 - (1 - 0.8289) = 0.6820.$$

**Interpretation.** 68.20% of commutes are between 30.5 and 50.5 minutes.

## 6 Problem 6: Normal Distribution Central Band with Both Negative z-bounds

**Problem.** Monthly sales follow  $X \sim N(120, 20^2)$  units. Find  $P(84 < X < 103)$ .



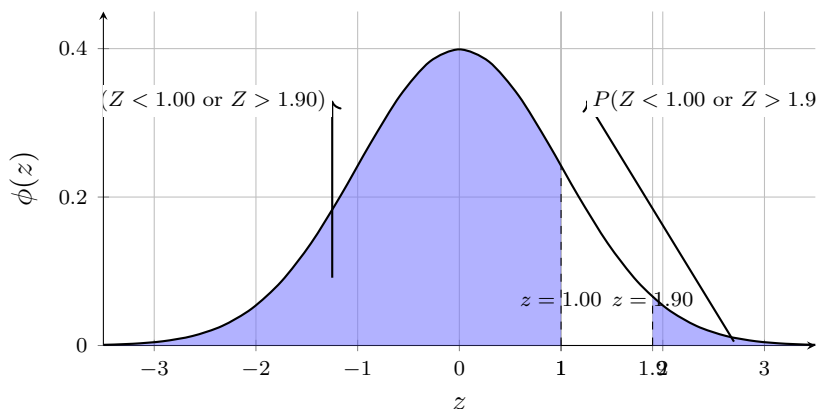
$$Z = \frac{X - 120}{20}, \quad P(84 < X < 103) = P(-1.80 < Z < -0.85).$$

$$P = \Phi(-0.85) - \Phi(-1.80) = (1 - 0.8023) - (1 - 0.9641) = 0.1618.$$

**Interpretation.** 16.18% of months have sales between 84 and 103 units.

## 7 Problem 7: Normal Distribution Two Regions with Positive z-bounds

**Problem.** Household electricity use follows  $X \sim N(400, 50^2)$  kWh. Find  $P(X < 450 \text{ or } X > 495)$ .



$$Z = \frac{X - 400}{50}, \quad P(X < 450 \text{ or } X > 495) = P(Z < 1.00 \text{ or } Z > 1.90).$$

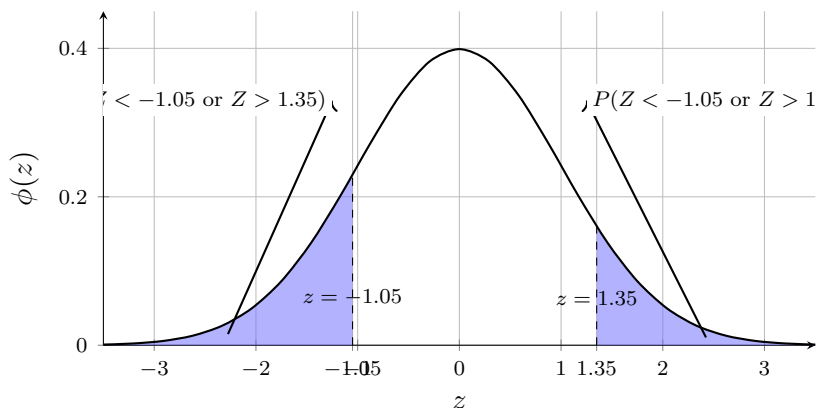
The regions are disjoint, so

$$P = \Phi(1.00) + (1 - \Phi(1.90)) = 0.8413 + 0.0287 = 0.8700.$$

**Interpretation.** 87.00% of homes are below 450 kWh or above 495 kWh.

## 8 Problem 8: Normal Distribution Two Tails with One Negative and One Positive z-bound

**Problem.** Scholarship scores follow  $X \sim N(300, 20^2)$ . Find  $P(X < 279 \text{ or } X > 327)$ .



$$Z = \frac{X - 300}{20}, \quad P(X < 279 \text{ or } X > 327) = P(Z < -1.05 \text{ or } Z > 1.35).$$

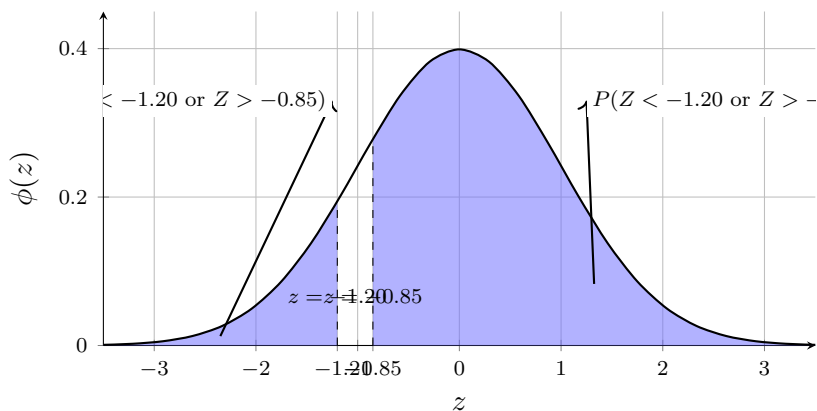
The regions are disjoint, so

$$P = \Phi(-1.05) + (1 - \Phi(1.35)) = (1 - 0.8531) + (1 - 0.9115) = 0.2354.$$

**Interpretation.** 23.54% are either below 279 or above 327 points.

## 9 Problem 9: Normal Distribution Two Regions with Both Negative z-bounds

**Problem.** Customer waiting times follow  $X \sim N(90, 20^2)$  seconds. Find  $P(X < 66 \text{ or } X > 73)$ .



$$Z = \frac{X - 90}{20}, \quad P(X < 66 \text{ or } X > 73) = P(Z < -1.20 \text{ or } Z > -0.85).$$

The regions are disjoint, so

$$P = \Phi(-1.20) + (1 - \Phi(-0.85)) = 0.1151 + 0.8023 = 0.9174.$$

**Interpretation.** 91.74% wait below 66 s or above 73 s.